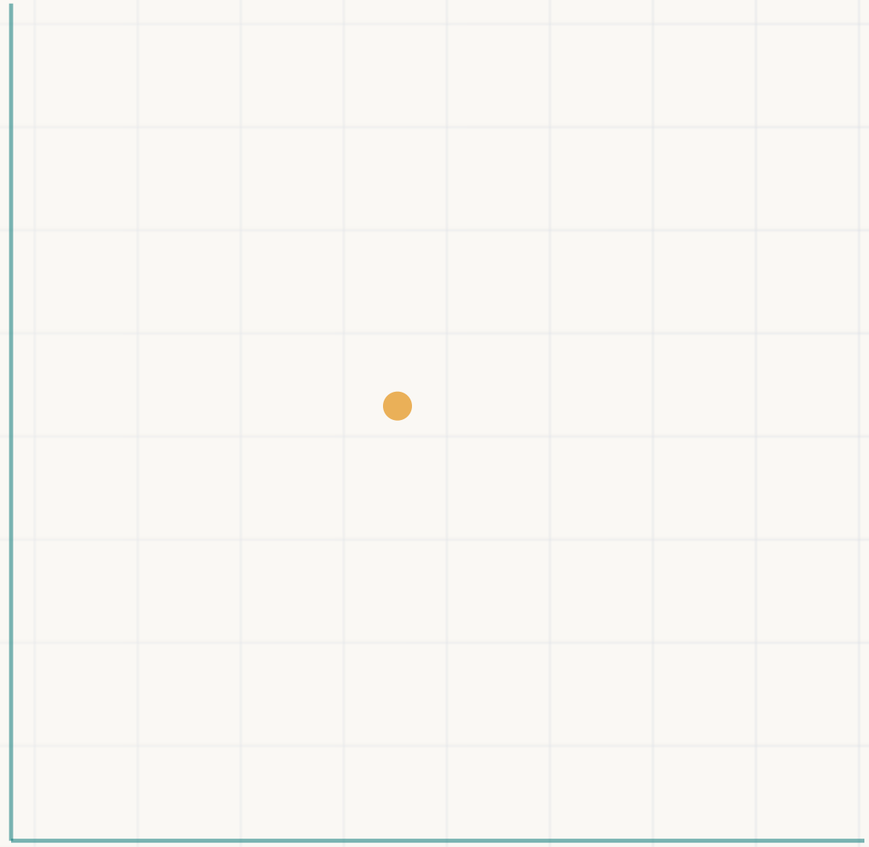
A 30-MINUTE ALGEBRA MICRO-LESSON

Linear Equations & Algebraic Modeling

From a city budget to a coordinate plane — **algebra is a language for describing relationships and constraints.**



Based on OpenStax Algebra 1 · Unit 1 Overview and Readiness

MIAMI'S BUDGET, AS AN EQUATION

A Budget Is a Constraint

$$I = E + R$$

I income · E expenses · R reserves

Teaching illustration — not data from the source text.

City Manager Arthur Noriega oversees Miami's budget.

“Algebra is a language for describing relationships and constraints. It helps us model real-world situations, solve problems, and make predictions.”

— OpenStax Algebra 1, Unit 1 Overview and Readiness (p. 37)

Variable

An unknown quantity, such as E.

Expression

Variables combined, such as $E + R$.

Equation

A relationship that must hold, such as $I = E + R$.

What Makes a Solution a Solution?

The solution of an equation is **the value of the variable that makes the equation a true statement.**

$$x + 2 = 3$$

$x = 1$ **is a solution — because it makes the equation true.**

CHECK: substitute $x = 1$ back into the original equation

$1 + 2 = 3 \rightarrow 3 = 3$ ✓

The Goal: Isolate the Variable

The goal is to **isolate the variable** — get it alone on one side of the equation.

$3x + 1 = 9 - x$ start — the variable appears on both sides

$4x + 1 = 9$ $+x$ to both sides

$4x = 8$ -1 from both sides

$x = 2$ divide both sides by 4

CHECK: substitute $x = 2$ back into the original equation

$3(2) + 1 = 9 - 2 \rightarrow 7 = 7$ ✓

A General Strategy in Five Steps

One strategy, five moves — every linear equation bends to it.

1

Simplify each side

Use the **Distributive Property** to remove parentheses. Combine like terms.

2

Collect variable terms

Move all variable terms to one side, using the Addition or Subtraction Property of Equality.

3

Collect constant terms

Move all constant terms to the other side, using the Addition or Subtraction Property of Equality.

4

Make the coefficient 1

Make the coefficient of the variable term equal to 1 — usually by dividing. State the solution.

5

Check the solution

Substitute it into the original equation and verify the statement is true.

Strategy in Action

Run the five-step strategy on a Try It problem: $2(3x - 8) = 19 - x$

1. $6x - 16 = 19 - x$ — distribute the 2 — the Distributive Property
2. $7x - 16 = 19$ — $+x$ to both sides
3. $7x = 35$ — +16 to both sides
4. $x = 5$ — divide both sides by 7

CHECK: substitute $x = 5$ back into the original equation

$2(3 \cdot 5 - 8) = 19 - 5 \rightarrow 2(7) = 14 \rightarrow 14 = 14$ ✓

Conditional: True for Some Values

A **conditional equation** is true for one or more values of the variable and false for others.

$$7x + 8 = -13$$ start

$$7x = -21$$ −8 from both sides

$$x = -3$$ ÷7

Conditional Solution: $x = -3$ — only this value makes the equation true.

Identity: True for Every Value

An **identity** is true for any value of the variable.

$$2y + 6 = 2(y + 3)$$ start

$$2y + 6 = 2y + 6$$ distribute the 2

$$6 = 6$$ $-2y$ from both sides — the variable vanishes

Identity All real numbers are solutions — the statement is true no matter what.

Contradiction: True for No Value

A **contradiction** is false for all values of the variable.

$$5z = 5z - 1$$ start

$$0 \neq -1$$ $-5z$ from both sides — the variable vanishes, leaving a false statement

Contradiction No solution — written as $\varnothing$ or “no solution”.

Three Kinds of Equations, One Test

Conditional

$7x + 8 = -13$

WHEN IT'S TRUE	True for one or more values, false for others
SOLUTIONS	**One or more values** (e.g., $x = -3$)

Identity

$2y + 6 = 2(y + 3)$

WHEN IT'S TRUE	True for any value
SOLUTIONS	**All real numbers**

Contradiction

$5z = 5z - 1$

WHEN IT'S TRUE	False for all values
SOLUTIONS	**No solution** ($\varnothing$)

One test for all three: simplify until the variable cancels or isolates, then read what's left — one value, every value, or no value.

Every Point Is an Ordered Pair

$$(x, y)$$

- first number — x-coordinate
- second number — y-coordinate

In the rectangular coordinate system, every point is represented by an **ordered pair**.

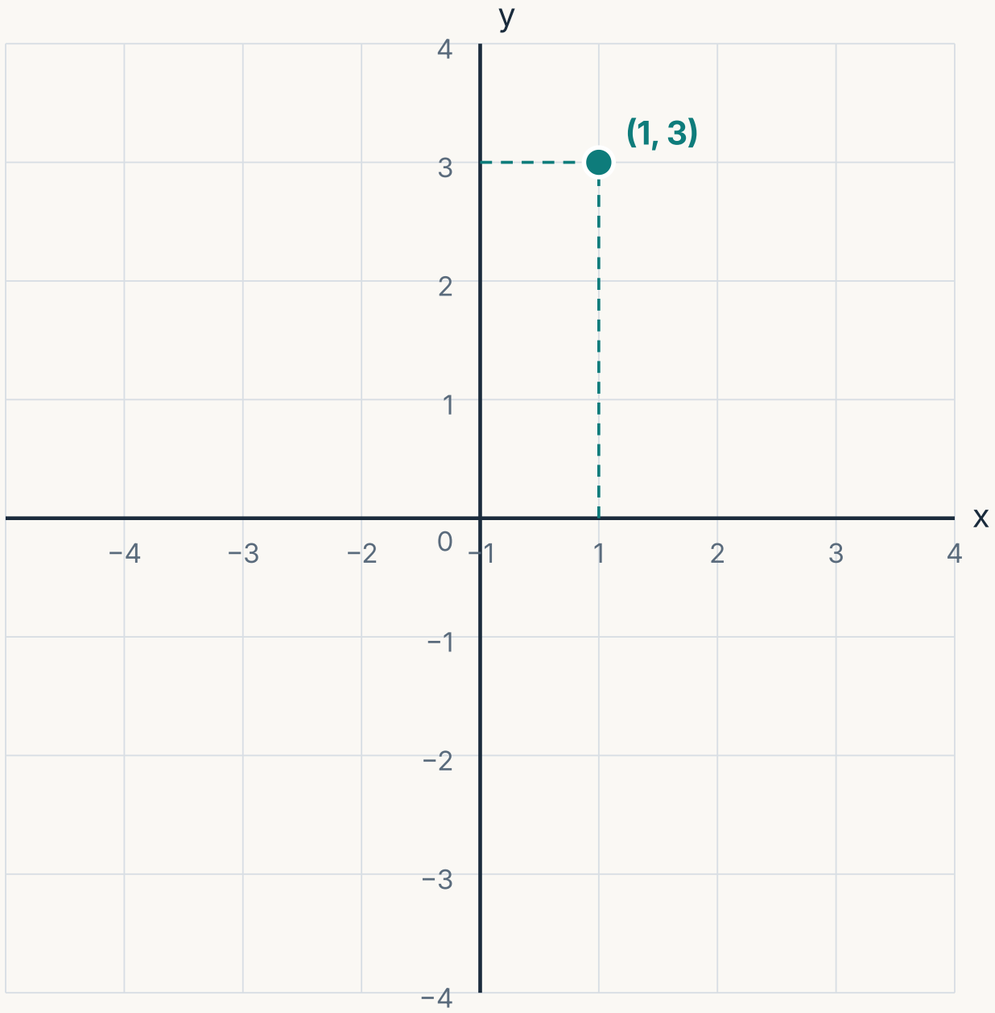
The phrase “ordered pair” means the order is important:

> **$(3, -4) \neq (-4, 3)$**



The point where the axes cross has ordered pair **(0, 0)** — called the **origin**.

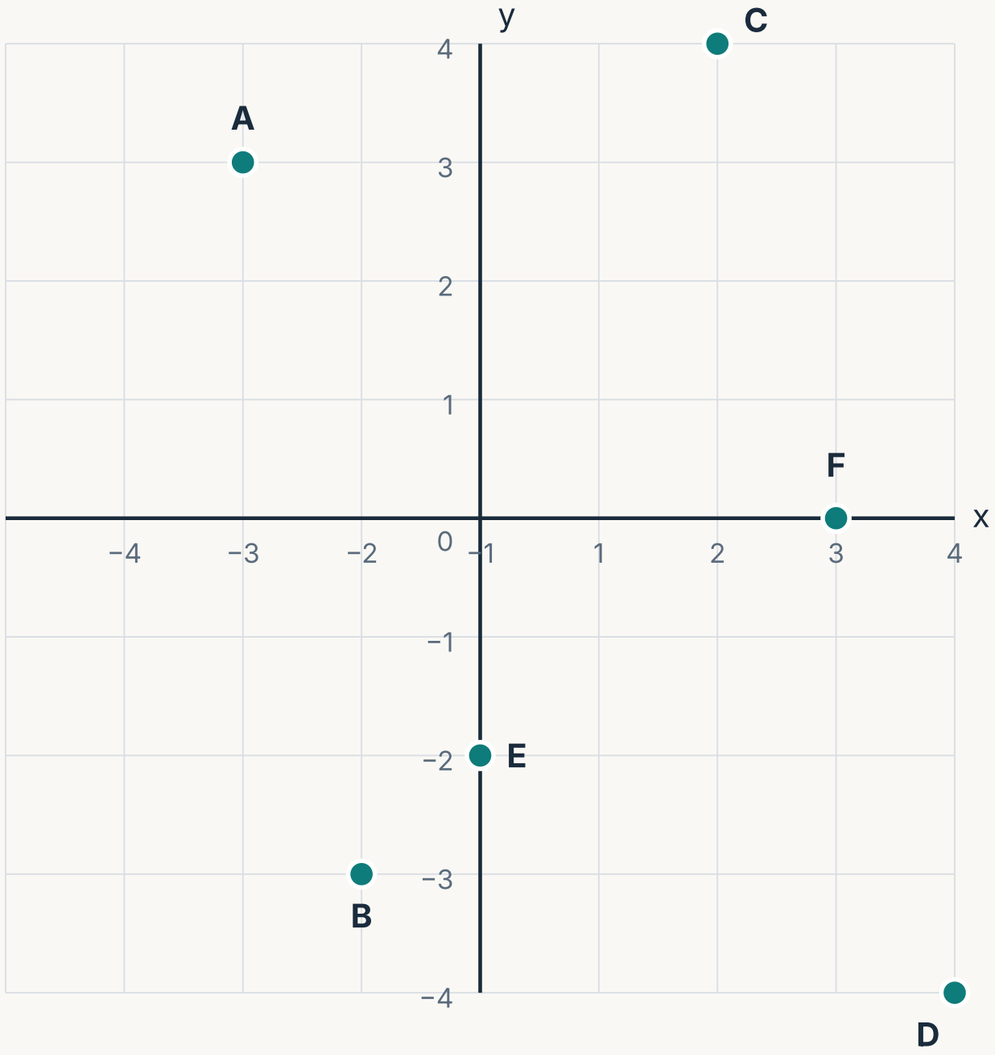
Plotting (1, 3): Where Two Lines Meet



1. Locate **1** on the x-axis; sketch a vertical line through $x = 1$.
2. Locate **3** on the y-axis; sketch a horizontal line through $y = 3$.
3. The point where the two lines meet is **(1, 3)**.

The guide lines are not part of the graph — they only help locate the point.

Reading Coordinates Off a Graph



Name the ordered pair of each point.

Read the **x**-coordinate on the **x**-axis directly above or below the point; read the **y**-coordinate on the **y**-axis directly to the left or right. Write the pair in the order **(x, y)**.

Answers

A (−3, 3) B (−1, −3) C (2, 4)
D (4, −4) E (0, −2) F (3, 0)

WRAP UP

Algebra Models the World

1. Solve linear equations using a general strategy.
2. Classify equations as conditional, identity, or contradiction.
3. Identify coordinates of points on a rectangular coordinate system.

> “Nobody is good at problems like this on the first try. **Keep at it, and it will get easier.**”

From the budget constraint $I = E + R$ to the point (1, 3) — algebra describes the relationships around us.